

# Md Omar Faruqe Riyad

 [Google Scholar](#) |  [riyad.omf@gmail.com](mailto:riyad.omf@gmail.com) |  [riyadomf](#) |  [riyadomf](#) |  [riyadomf.github.io](#)

Research interests: AI agents for site reliability engineering and software engineering; NLP for low-resource languages.

## EDUCATION

### Shahjalal University of Science and Technology (SUST)

Sylhet, Bangladesh

*B.Sc. in Computer Science and Engineering; CGPA: 3.53/4.00 (Last two years: 3.83)*

2018 – 2023

- GRE: 317/340 · TOEFL: 104/120

## RESEARCH EXPERIENCE

### AI for Site Reliability Engineering (SREGym) | *Research Intern, advised by Prof. Tianyin Xu*

UIUC, 2026

- Working on the SREGym v2 paper: extending [SREGym](#), an open-source benchmark for AI SRE agents, from synthetic faults to real-world failure scenarios, and analyzing how agents behave on them.
- Turning documented production outages into reproducible benchmark problems through case studies and postmortem analysis, then designing fault scenarios and oracles that target known agent weaknesses: cross-layer reasoning gaps, greedy diagnosis anchoring, and reward hacking.

### Code Generation for Bangla | *BLP-2025 Shared Task 2, first author*

2025

- Fine-tuned Llama 3, TigerLLM, and Qwen with LoRA adapters for Bangla-to-Python code generation, a setting with little supervised data and no strong open baseline.
- Added Chain-of-Thought decomposition and a self-refinement loop that repairs generated code from execution feedback, reaching **Pass@1 0.85**; categorized failures by type (syntax, runtime, misread problem statement) to steer refinement.

### Violence Inciting Text Detection in Bangla | *BLP-2023 Shared Task 1, first author*

2023

- Countered class imbalance and data scarcity with semi-supervised self-training over unlabeled Bangla text plus back-translation augmentation, then ensembled fine-tuned Bangla transformers with bagging and majority voting to cut variance on minority classes.

### Visual Question Answering (VQA) Data Synthesis | *Undergraduate Thesis*

2022

- Augmented VQA v2 training data with automatically generated question-answer pairs to widen answer-space diversity, then measured where models fall back on answer-frequency bias instead of visual reasoning.

## PUBLICATIONS

- [1] **Omar Faruqe Riyad** and Jahedul Alam Junaed. [AdversaryAI at BLP-2025 Task 2: A Think, Refine, and Generate \(TriGen\) System with LoRA and Self-Refinement for Code Generation](#). BLP Workshop at IJCNLP-AAACL 2025.
- [2] Trina Chakraborty, Marufur Rahman, and **Omar Riyad**. [Deja Vu at SemEval 2024 Task 9: A Comparative Study of Advanced Language Models for Commonsense Reasoning](#). SemEval Workshop at NAACL 2024.
- [3] **Omar Faruqe Riyad**, Trina Chakraborty, and Abhishek Dey. [Team\\_Syrax at BLP-2023 Task 1: Data Augmentation and Ensemble Based Approach for Violence Inciting Text Detection in Bangla](#). BLP Workshop at EMNLP 2023.

## EXPERIENCE

### Dynamic Solution Innovators

April 2023 – Present

*Software Engineer*

*Dhaka, Bangladesh*

- Contributed to the data pipeline behind the voter registration flow of EC-BVRS, Bangladesh's national voter registration (NID) platform, which processes records for nearly **150 million voters** (Spring Boot, Kafka, Redis).
- Built and open-sourced [gf](#) , a GlassFish CLI that hot-swaps modified bytecode into running JVMs over JDI/JDWP and bypasses JasperReports classloader caching. Cut local edit-deploy cycles from **~2 minutes to ~5 seconds**.
- Introduced Prometheus, Loki, and Grafana observability for GlassFish applications, exposing JVM memory, thread pools, and request latency for production debugging. Also serve on the internal Cybersecurity Team, testing releases against OWASP Top 10 risks.

### Altri.ai

*Full-Stack AI Engineer, RevestAI *

Part-time

*Remote, US*

- Designed the three-stage property scoring pipeline that separates LLM feature extraction, ML valuation, and deterministic financial heuristics, so each layer can be retrained and evaluated on its own.
- Rewrote the sequential LLM enrichment pipeline as bounded-concurrency async batches with atomic per-batch commits. Throughput rose about **10×**, and failed runs resume without repeating completed LLM calls.
- Implemented distributed job coordination on PostgreSQL with `SELECT FOR UPDATE` row locks and stale-lock recovery, so multiple scheduler replicas safely share long-running enrichment and training jobs.
- Built the property ingestion pipeline on PostgreSQL with content-hash deduplication and incremental delta-syncing, so repeated scrapes skip already-enriched listings while preserving historical state. Model promotion is gated the same way: a challenger must beat the champion on a gold-standard holdout slice, and a human signs off before production-wide rescoreing.

## PROJECTS

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>SREGym Fault Scenario Contribution</b>    <i>Python, Kubernetes, Docker</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2026 |
| <ul style="list-style-type: none"> <li>• Reproduced Kubernetes <a href="#">issue #128162</a> as a merged benchmark scenario: cumulative mutating-webhook timeouts exceed the API server's admission deadline and silently block all pod creation, with no crash to signal the fault.</li> <li>• Designed a four-property mitigation oracle ending in a fresh probe-pod admission test, which rejects reward-hacking fixes such as deleting all webhooks that mask the symptom without restoring the control-plane network path.</li> </ul>                                                                                                                                                                                                                                                                                                                                               |      |
| <b>Token-Efficient LLM Query Router</b>    <i>AMD Hackathon Act II, Track 1</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2026 |
| <ul style="list-style-type: none"> <li>• Designed a four-tier inference cascade (regex router → ONNX classifier → LoRA fine-tuned Qwen2.5-1.5B → cloud LLM fallback) that resolves 8 task categories with minimal paid API usage, admitting each tier only at measured precision (router 64/64 on adversarial cases; sentiment classifier 100%).</li> <li>• Replaced model confidence checks with deterministic verification guards: sandboxed AST evaluation of model-emitted math expressions, which removes arithmetic errors and cuts output tokens ~90%, and an entity-completeness check that catches 99% of dropped entities in NER.</li> <li>• Fine-tuned the local tier on 2,800 programmatically validated synthetic examples and iterated 17 times against an LLM-as-judge harness mirroring the competition rubric, root-causing each regression before redeploy.</li> </ul> |      |
| <b>KnowledgeRelay</b>    <i>Python, FastAPI, LangChain, ChromaDB, React</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2025 |
| <ul style="list-style-type: none"> <li>• Built a knowledge-transfer agent that interviews outgoing team members with adaptive questions, ingests their documents and code, and answers new joiners' questions with source attribution. Retrieval rewrites follow-ups into self-contained queries.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |      |
| <b>PrimeFaces Open Source Contribution</b>    <i>Java, JSF, PrimeFaces</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2024 |
| <ul style="list-style-type: none"> <li>• Contributed a feature enhancement to PrimeFaces' <code>JPALazyDataModel</code> that lets developers inject custom filters into DataTable-backed JPA queries.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |      |

## HONORS, AWARDS & CERTIFICATIONS

|                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>4th place</b> out of 32 teams, Code Generation in Bangla Shared Task, BLP at IJCNLP-AAACL 2025 <a href="#">[Leaderboard]</a>         |
| <b>6th place</b> out of 30 teams, DSI AI Agent Hackathon 2025 <a href="#">[Leaderboard]</a>                                             |
| <b>12th place</b> in post-evaluation (19th official), Violence Inciting Text Detection, BLP at EMNLP 2023 <a href="#">[Leaderboard]</a> |
| <b>11th place</b> , BUET CSE Fest 2022 AI Contest                                                                                       |
| <b>Certified in Cybersecurity (CC)</b> , ISC2 <a href="#">[Badge]</a>                                                                   |

## TEACHING & SERVICE

|                                                                                                                                                                                                                                                                                                                   |                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>Course Instructor</b>                                                                                                                                                                                                                                                                                          | 2024 – Present |
| <i>Dynamic Learning</i>                                                                                                                                                                                                                                                                                           | <i>Remote</i>  |
| <ul style="list-style-type: none"> <li>• Authored and taught <a href="#">Enterprise Web Development with JEE, JSF, and PrimeFaces</a>: component-driven design, CDI/EJB integration, and JSF lifecycle management. </li> </ul> |                |
| <b>Volunteer</b> , IEEE International Conference on Bangla Speech and Language Processing (ICBSLP)                                                                                                                                                                                                                | 2019           |

## TECHNICAL SKILLS

|                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------|
| <b>Languages:</b> Java, Python, SQL, Bash, JavaScript, C/C++                                                            |
| <b>Backend &amp; Systems:</b> Jakarta EE (JSF, CDI, EJB, JPA, JMS), Spring Boot, FastAPI, Kafka, GlassFish              |
| <b>AI &amp; ML:</b> PyTorch, LLMs, LoRA fine-tuning, RAG, LangChain, XGBoost                                            |
| <b>Data &amp; Infrastructure:</b> PostgreSQL, MySQL, Redis, MinIO, Docker, Kubernetes, Prometheus, Grafana, Loki, Nginx |
| <b>Security:</b> OWASP Top 10, RBAC, JWT                                                                                |